

Microphone STT Benchmark Report

Speech-to-Text Accuracy Analysis Across Microphone Types

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17 Microphone Samples Evaluated

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1 Executive Summary

This report presents the results of a comprehensive benchmark comparing speech-to-text (STT) accuracy across 17 different microphone samples representing various types, price points, and use cases. All samples were recorded reading the same reference text about the history of coffee, then transcribed using OpenAI’s Whisper API to ensure methodological consistency.

Key Findings:

- The **cheapest microphone** (UGreen gooseneck, \$18) achieved the **best** Word Error Rate (4.44%)
- Price and accuracy show a **positive correlation** ($r=0.578$, $p=0.039$) — meaning expensive microphones tend to perform **worse**
- USB Speakerphones offer excellent value, with the EMEET (\$40) outperforming the Jabra Speak 510 (\$110)
- Dedicated digital voice recorders don’t necessarily excel at STT tasks

2 Methodology

2.1 Recording Setup

All recordings were made in the same environment:

- **Location:** Home office, quiet acoustic environment
- **Recording Software:** Audacity (default settings)
- **Gain:** 100% (unless otherwise noted)
- **Reference Text:** 315-word passage about the history of coffee
- **Audio Format:** WAV (44.1kHz, 16-bit) or MP3 where noted

2.2 Evaluation Metrics

Metric	Description
Word Error Rate (WER)	Percentage of words incorrectly transcribed (lower is better)
Character Error Rate (CER)	Percentage of characters incorrectly transcribed
Audio Quality Score	Composite score (0-100) based on SNR, dynamic range, clipping, etc.
Signal-to-Noise Ratio (SNR)	Measured in dB; higher indicates cleaner audio

2.3 Transcription Service

All samples were transcribed using **OpenAI Whisper API (whisper-1)** to ensure methodological consistency. Using a single transcription service eliminates variability from different STT engines.

3 Results

3.1 Overall Rankings by Word Error Rate

The following table shows all 17 microphone samples ranked from best (lowest WER) to worst:

Rank	Microphone	Type	Price	WER	Quality
1	OnePlus Nord 3 5G	Smartphone	—	4.13%	79.5
2	UGreen CM564	USB Gooseneck	\$18	4.44%	70.0
3	Audio-Technica ATR4697	Boundary Mic	\$45	4.76%	57.0
4	EMEET Conference	USB Speakerphone	\$40	5.08%	75.0
5	Samson Q2U	Dynamic USB/XLR	\$70	5.40%	80.0
6	OnePlus Nord 3 5G	Smartphone (MP3)	—	5.40%	83.0
7	Audio-Technica ATR4697	Boundary (80cm)	\$45	5.40%	70.0
8	Jabra Speak 510	USB Speakerphone	\$110	5.40%	80.0
9	Logitech C925e	Webcam	\$75	5.40%	75.0
10	Maono Elf	Lavalier	\$25	5.40%	75.0
11	Logitech H390	USB Headset	\$30	5.71%	74.0
12	OnePlus Nord 3 5G	Smartphone	—	5.71%	84.5
13	Yealink BH72	Wireless Headset	\$160	6.03%	85.0
14	Yealink BH72	Wireless (BT)	\$160	6.03%	85.0
15	OnePlus Nord 3 5G	Smartphone (noisy)	—	6.03%	89.0
16	Sony ICD-UX570	Digital Voice Rec.	\$100	6.03%	78.0
17	Audio-Technica ATR4750	Condenser Gooseneck	\$55	6.35%	70.0

3.2 Rankings by Microphone Type

When grouped by microphone type, clear patterns emerge:

Type	Avg WER	Range	Avg Price
USB Gooseneck	4.44%	4.44%	\$18
Boundary Microphone	5.08%	4.76–5.40%	\$45
USB Speakerphone	5.24%	5.08–5.40%	\$75
Smartphone built-in	5.32%	4.13–6.03%	N/A
Dynamic USB/XLR	5.40%	5.40%	\$70
Webcam built-in	5.40%	5.40%	\$75
Lavalier	5.40%	5.40%	\$25
USB Headset	5.71%	5.71%	\$30
Wireless Headset	6.03%	6.03%	\$160
Digital Voice Recorder	6.03%	6.03%	\$100
Condenser Gooseneck	6.35%	6.35%	\$55

Key observations:

- Simple USB gooseneck microphones perform best on average
- Expensive wireless headsets rank near the bottom
- Conference speakerphones offer competitive accuracy at reasonable prices
- Smartphone microphones show high variability depending on environment

4 Price-Performance Analysis

4.1 Correlation Analysis

Statistical Finding: There is a statistically significant **positive** correlation between price and Word Error Rate.

Pearson r	0.578
p-value	0.039
Interpretation	Higher price → Higher error rate

This counterintuitive result suggests that **more expensive microphones do not necessarily provide better STT accuracy**.

4.2 Head-to-Head Comparisons

4.2.1 Conference Speakerphones

Model	Price	WER	Value
EMEET Conference	\$40	5.08%	Better
Jabra Speak 510	\$110	5.40%	—

The EMEET is **2.75× cheaper** than the Jabra while achieving **better** accuracy. This represents exceptional value for conference/speakerphone use cases.

4.2.2 Gooseneck Microphones

Model	Price	WER	Type
UGreen CM564	\$18	4.44%	USB Gooseneck
Audio-Technica ATR4750	\$55	6.35%	Condenser Gooseneck

The budget UGreen significantly outperforms the more expensive Audio-Technica condenser gooseneck.

4.2.3 Best Value Overall

Best Value: UGreen CM564 USB Gooseneck

- Price: \$18
- WER: 4.44%
- WER per \$10 spent: 2.47%

This microphone offers the lowest error rate at the lowest price point among all tested devices.

5 Detailed Audio Metrics

5.1 Sample Audio Characteristics

ID	Microphone	SNR (dB)	Peak (dB)	RMS (dB)	Quality
1	UGreen CM564	15.2	-1.8	-25.1	70.0
2	Samson Q2U	25.2	-3.2	-24.9	80.0
3	Logitech H390	21.4	-0.8	-28.3	74.0
4	OnePlus Nord 3 5G	26.1	-0.9	-19.0	84.5
6	AT ATR4697 (30cm)	11.4	-6.9	-29.1	70.0
7	AT ATR4697 (80cm)	8.1	-10.8	-32.4	57.0
8	Jabra Speak 510	24.1	0.0	-18.6	80.0
9	Logitech C925e	16.6	-2.1	-24.2	75.0
10	Maono Elf	20.5	0.0	-22.8	75.0
11	Yealink BH72 (dongle)	37.2	-2.9	-23.8	85.0
12	Yealink BH72 (BT)	35.2	-2.8	-24.2	85.0
13	AT ATR4750-USB	17.9	-5.7	-25.1	70.0
14	OnePlus (noisy)	35.7	-0.8	-13.7	89.0
15	OnePlus (quiet)	24.8	-0.9	-21.3	79.5
16	EMEET Conference	25.6	-5.1	-27.3	75.0
17	Sony ICD-UX570	31.5	0.0	-21.6	78.0

Note: Higher Audio Quality scores don't always correlate with better STT accuracy. The Yealink BH72 has the highest quality score (85.0) but one of the worst WERs (6.03%).

6 Conclusions and Recommendations

6.1 Key Takeaways

1. **Price doesn't predict accuracy.** The statistical correlation shows expensive microphones actually tend to perform worse for STT.
2. **Simple is often better.** Basic USB gooseneck microphones outperform sophisticated wireless headsets and dedicated voice recorders.
3. **Conference speakerphones offer great value.** The EMEET at \$40 provides professional-grade STT accuracy comparable to microphones costing 2-3× more.
4. **Smartphones are surprisingly capable.** In quiet environments, smartphone microphones perform competitively with dedicated hardware.
5. **Audio quality ≠ STT accuracy.** High SNR and quality scores don't guarantee better transcription results.

6.2 Recommendations by Use Case

Use Case	Recommendation	Budget
Best overall value	UGreen CM564 USB Gooseneck	\$18
Conference calls	EMEET Conference Speakerphone	\$40
Professional recording	Samson Q2U	\$70
Portable/travel	Smartphone (quiet environment)	N/A
Hands-free dictation	Logitech H390 USB Headset	\$30

6.3 Limitations

- All tests conducted in a single environment
- Single speaker (may not generalize across voice types)
- Single STT engine (OpenAI Whisper)
- Reference text is English only

6.4 Future Work

- Test additional microphone types (shotgun, XLR condensers)
- Compare multiple STT engines
- Test in varied acoustic environments
- Evaluate with multiple speakers

Report generated: February 04, 2026

Repository: <https://github.com/danielrosehill/Microphone-Audio-Samples>

Methodology: OpenAI Whisper API (whisper-1), single-pass transcription